Practice workbook · The internet, the web, DNS and the parts of a URL

What the internet and WWW are, how DNS resolves names, and breaking a URL into protocol, domain, folder and file or classifying URL/domain/IP items.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-11

1. Question 3 · part b.iv · 4 marks

2022 JUN · Paper 01

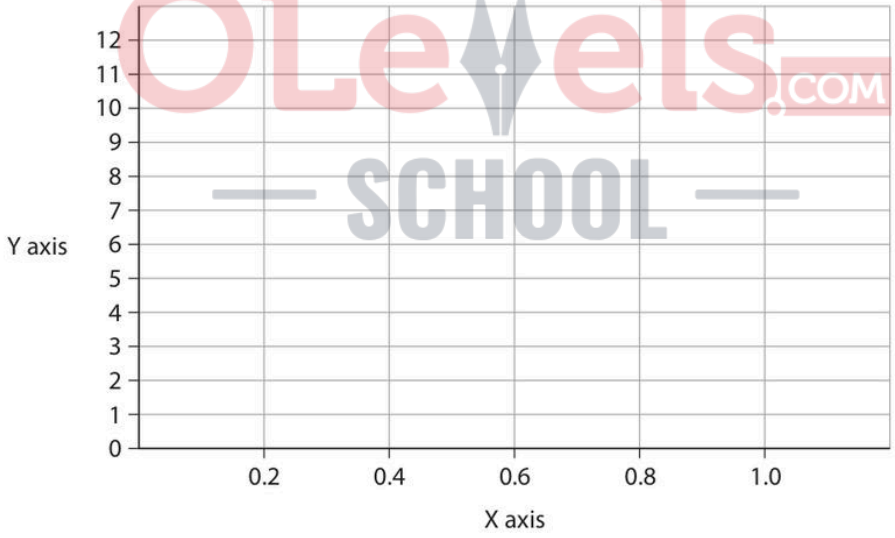
Worked-example retry · Attempt part b.iv only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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8

P 7 2 5 3 8 A 0 8 2 0

1. Question 3 · part b.iv · 4 marks

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Worked-example retry · Attempt part b.iv only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

1. Question 3 · part b.iv · 4 marks · Working and answer

Working and answer

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2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	Zafer
Washing machine design drawings		
A folder of manuals for new machines		

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P 6 1 8 8 0 A 0 4 2 4

2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

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2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

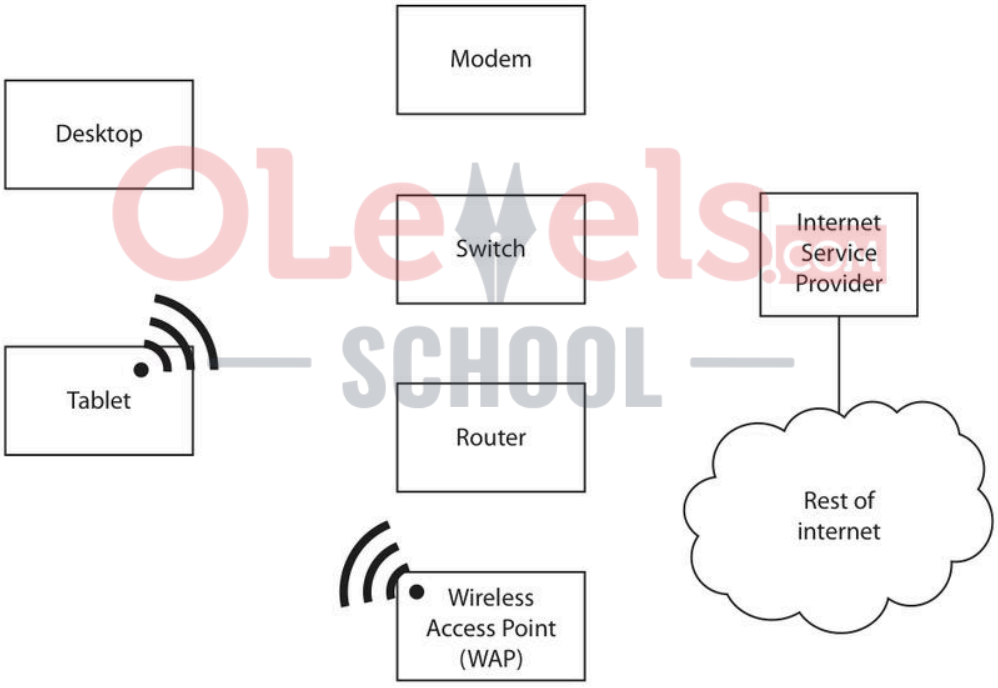
(d) Zafer can use a desktop or a tablet computer to connect to the internet.

Only the tablet has wireless capabilities.

The diagram shows the components used to connect to the internet.

Complete the diagram to show how the desktop and the tablet are connected to the internet.

(6)



(Total for Question 2 = 12 marks)

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2. Question 2 · part c · 2 marks · Working and answer

Working and answer

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3. Question 1 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

3. Question 1 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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3. Question 1 · part c · 2 marks · Working and answer

Working and answer

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4. Question 2 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

4. Question 2 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

(e) **Figure 2** shows a network topology.

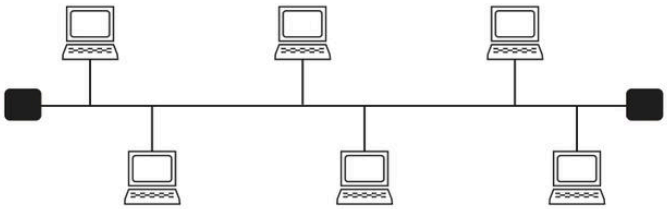


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1

2

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P 6 6 4 8 9 A 0 6 2 0

4. Question 2 · part d · 3 marks · Working and answer

Working and answer

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5. Question 1 · part e.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.i only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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5. Question 1 · part e.i · 1 mark

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Further practice · Attempt part e.i only.

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(c) **Figure 1** shows a network topology.

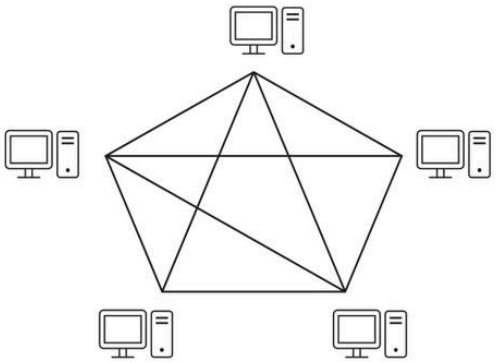


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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5. Question 1 · part e.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.i only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

.....

(ii) Give the name of the document being retrieved.

(1)

.....

(Total for Question 1 = 10 marks)

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Working and answer

6. Question 1 · part e.ii · 1 mark

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Further practice · Attempt part e.ii only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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6. Question 1 · part e.ii · 1 mark

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Further practice · Attempt part e.ii only.

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(c) **Figure 1** shows a network topology.

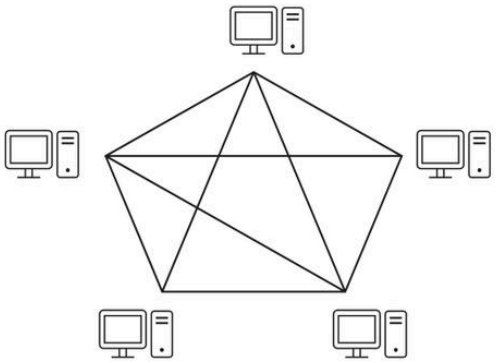


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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6. Question 1 · part e.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part e.ii only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

(Total for Question 1 = 10 marks)

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Working and answer

7. Question 1 · part a · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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7. Question 1 · part a · 4 marks · Working and answer

Working and answer

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